ARJUN GOPAL

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Professional Summary

AI/ML engineer and M.Tech candidate specialising in Artificial Intelligence, Deep Reinforcement Learning, and Neural Reinforcement Optimisation. Experienced in full-stack development (Django, REST APIs) and ML workflows. Strong foundation in algorithms, data structures, and systems — seeking to tackle large-scale, computationally complex problems.

Education

Amrita Vishwa Vidyapeetham **Coimbatore, India**
M.Tech — Computer Science & Engineering (AI) *July 2025 – July 2027*

- Coursework focus: Artificial Intelligence, Robotics, Deep Learning, Advanced ML Systems.

Sathyabama Institute of Science & Technology **Chennai, India**
B.E. — Artificial Intelligence & Machine Learning (cgpa:9.57) *2021 – 2025*

- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, Operating Systems, DBMS.
- Certifications: REDHAT System Administrator(RHCSA),Data Analysis for Machine Learning(IBM), MATLAB Machine Learning, GATE 5k rank,Oracle Certified Foundation Associate, Digital 101.

Experience

Virtusa **India**
Software Engineering Intern *July 2023 – February 2024*

- Designed and deployed RESTful web applications using **Django** (Python) backend framework, serving dynamic content with efficient ORM queries and MVC architecture.
- Implemented responsive front-end interfaces leveraging **Bootstrap**, HTML5, CSS3, and JavaScript, improving cross-device compatibility across projects.
- Collaborated in an Agile team environment, participating in code reviews, sprint planning, and iterative feature delivery across an 8-month engagement.

HCLTech **Chennai, India**
Software Engineering Intern (Student) *July 2023 – November 2023*

- Built an interactive **art gallery web application** from scratch using HTML5, CSS3, and vanilla JavaScript, focusing on performance, accessibility, and visual fidelity.
- Applied UI/UX principles to enhance user engagement, delivering a scalable single-page experience optimised for multiple screen resolutions.

Academic & Personal Projects

Emotion-Based Music Genre Classification *2024 – 2025*

- Fine-tuned a pretrained VGG-16 CNN on a custom emotion-labelled dataset using transfer learning and domain adaptation; built an end-to-end preprocessing pipeline covering face detection, landmark extraction, and class-weighted loss to handle imbalanced emotion categories.
- Mapped predicted emotion outputs to curated genre playlists, delivering a context-aware music recommendation system; evaluated with per-class F1 scores and confusion matrix analysis across all emotion-genre label pairs.

Driver Drowsiness Detection *2026*

- Built a real-time drowsiness detection system using sequential architectures (RNN, LSTM, GRU) to classify alert vs. drowsy states from temporal Eye Aspect Ratio (EAR) and head-pose feature sequences; benchmarked all three models — GRU achieved the best accuracy-latency trade-off over long sequences due to superior gradient flow.
- Deployed the best-performing model to Hugging Face Spaces with a Gradio UI for browser-based real-time inference; integrated an alert trigger mechanism requiring zero local setup from end users.

Technical Skills

Languages	Python, JavaScript, HTML5, CSS3, SQL
ML / AI	Deep Reinforcement Learning, Quantum Computing, Genetic Algorithms, Neural Networks, Sequential Networks, PyTorch, MATLAB
Frameworks	Django, Bootstrap, REST APIs
Tools	Git, GitHub, Linux, VS Code, Jupyter Notebook
Concepts	Data Structures & Algorithms, OOP, Agile/Scrum, MVC Architecture

Achievements & Leadership

- **Cultural Event Representative** — ASE 2026, Amrita Vishwa Vidyapeetham; helped incoordinating large-scale inter-collegiate cultural events.
- **Marketing Event — 2nd & 3rd Place**: Demonstrated strategic communication, competitive analysis, and persuasive presentation skills in an inter-collegiate business challenge held at **NIT Trichy.**
- **RHCSA**(Red Hat Certified System Administrator) — Red Hat; validates hands-on proficiency in core Linux system administration including file systems, user/group management, process control, networking, SELinux, and shell scripting in a live exam environment..
- Qualified **GATE** (Graduate Aptitude Test in Engineering), validating strong command of CS fundamentals including algorithms, data structures, and theory of computation.
- **Navy Cadet (National Cadet Corps)** : Commissioned as a Navy Cadet under the NCC during undergraduate studies; participated in naval training camps developing strong leadership, teamwork, and crisis-management skills.